1. Scope and course design basis

This review replaces the provisional 120 W/m2 design with the lecturer's Week 8 workshop basis. It covers supply diffuser selection at maximum and 40% flow, VAV grouping, flexible routes and rectangular supply duct sizing. Original architectural files are preserved.

Parameter	Adopted value	Basis
Office load area	1022 m2	Printed architectural plan
Total cooling allowance	189 W/m2	Workshop p.1 office row
Sensible heat factor	0.90	Workshop p.1 office guidance
Room / supply air	24 / 12 deg C	Declared design assumption
Ceiling height	2.7 m	Declared design assumption
Minimum airflow	40%	Assignment task 2
Course T0.25/L range	1.0-3.4	Workshop p.5 model 1400 example
Main / branch velocity	10 / 8 m/s maximum	Workshop p.7 office row
Rigid friction limit	0.8 Pa/m	Chosen within course 0.65-1.23 range
Rectangular aspect	No greater than 4:1	Assignment preferred limit
Flexible length limit	4 m	FSD 2/2023 s.7.2; brief allows 5 m

Workshop: Space air diffusion design and air duct sizing. Read from Canvas PDF preview; page 5 table and example also visually checked. A local original PDF download was not obtained.

Course-method qualification. The workshop explicitly applies the louvered 1.0-3.4 range to model 1400. Manufacturer engineering guidance instead identifies a Type 3 circular pattern. Also, the reproduced table band is 35-160 W/m2, whereas the adopted sensible allowance is 170.1 W/m2. Results here demonstrate the stated course calculation method, not measured ADPI or independent manufacturer certification. This discrepancy is retained transparently.

This remains a review issue pending final drawing coordination and the remaining verification gates. It is not a construction specification or an assertion of full statutory compliance.

2. Load and airflow calculations

Total cooling = 1022 x 189 = **193.158 kW**. Sensible cooling = total x 0.90 = **173.842 kW**. These are thermal loads, not electrical power.

Q = sensible load / [rho x cp x (Tr - Ts)]. Using rho = 1.2 kg/m3, cp = 1006 J/(kg K), and delta T = 12 K:

Qmax = **12000.4 L/s**; Qmin = 0.40 Qmax = **4800.1 L/s**.

The rectangular layout cells total 1042.10 m2, including small columns/recesses. Their relative areas allocate the given 1022 m2; the envelope is not claimed as measured net area. Each outlet airflow equals total flow times allocated area / 1022. VAV flows are sums of their four outlet flows.

VAV	Outlets	Maximum L/s	Minimum L/s
VAV-01	4	603.7	241.5
VAV-02	4	579.6	231.8
VAV-03	4	603.7	241.5
VAV-04	4	603.7	241.5
VAV-05	4	579.6	231.8
VAV-06	4	623.0	249.2
VAV-07	4	570.0	228.0
VAV-08	4	547.2	218.9
VAV-09	4	570.0	228.0
VAV-10	4	570.0	228.0
VAV-11	4	547.2	218.9
VAV-12	4	588.3	235.3
VAV-13	4	595.9	238.4
VAV-14	4	572.1	228.8
VAV-15	4	595.9	238.4
VAV-16	4	595.9	238.4
VAV-17	4	572.1	228.8
VAV-18	4	615.0	246.0
VAV-19	4	733.7	293.5
VAV-20	4	733.7	293.5

3. Diffuser selection and checks

80 Krueger 1400/51400 four-cone diffusers are selected, with 600 x 600 mm panels and 8-inch round necks. Neck connection transitions are required from the nominal 250 mm flex. Catalogue basis: no damper, horizontal isothermal throw, B1-12 metric rows. The third throw value is T0.25.

Within adjacent catalogue rows, T = T1 + (T2-T1)(Q-Q1)/(Q2-Q1). Both operating flows must lie within the tabulated range; no extrapolation is used. Static pressure and NC are approximate interpolations. A catalogue dash denotes less than NC10, not measured zero.

Characteristic lengths are checked in four horizontal directions to adjacent walls or the midplane between opposing outlets. Across the open L-shaped junction, the adjacent row midplane replaces the artificial zone boundary. Reporting the smallest and largest directional T/L values makes edge conditions visible.

All 80 outlets satisfy the adopted course 1.0-3.4 criterion at both maximum and 40% airflow. The highest interpolated catalogue NC is 34.9. This is a catalogue sound criterion estimate with its stated room-absorption assumption, not a prediction of installed whole-room noise.

Worked example SD-01	Maximum	40% flow
Airflow L/s	152.25	60.90
T0.25 m	4.445	2.684
Directional ratio minimum	2.117	1.278
Directional ratio maximum	2.796	1.688

The workshop suggests 50-150 L/s as an approximate initial outlet-count guide. This is not a product rating limit: some selected outlets exceed 150 L/s, including approximately 217 L/s in the upper zone. Their throws, catalogue pressure and sound data remain within the published product range.

The eight upper-zone outlets use equal spacing to remove the short opposing-jet distance that caused two candidate failures. These positions require local coordination of the 600 mm ceiling grid. Final ceiling/beam levels remain assumptions.

4. VAV and flexible connections

20 VAV boxes each serve four outlets. Each 1000 x 800 mm box is a schematic layout envelope, not an identified manufacturer model. Direct flexible routes include 0.50 m allowance for connections and vertical movement. Nominal flexible duct diameter is 250 mm.

VAV	Longest design flex m	Connections
VAV-01	2.410	4
VAV-02	2.410	4
VAV-03	2.410	4
VAV-04	2.410	4
VAV-05	2.410	4
VAV-06	2.410	4
VAV-07	2.203	4
VAV-08	2.203	4
VAV-09	2.203	4
VAV-10	2.203	4
VAV-11	2.203	4
VAV-12	2.203	4
VAV-13	2.410	4
VAV-14	2.410	4
VAV-15	2.410	4
VAV-16	2.410	4
VAV-17	2.410	4
VAV-18	2.410	4
VAV-19	2.607	4
VAV-20	2.607	4

Maximum design flex length = 2.607 m, within the adopted 4 m limit and the assignment's 5 m limit. Actual flex must be fully extended and installed with minimum bends; excess compression and fitting losses are not covered by the rigid straight-duct calculation.

FSD Circular 2/2023 sections 7.1-7.2 govern material/test requirements and normal flex length. The AHU/corridor crossing requires a coordinated fire-enclosure and penetration-protection strategy (sections 3 and 9.2). Symbols alone do not demonstrate fire compliance.

5. Supply duct sizing method

The single-line tree has 49 rigid sections. Main flow starts at 12000.4 L/s; each branch carries the sum of downstream VAV maximum flows. Vertical distribution runs beside VAV envelopes, with short take-offs, rather than through boxes serving other branches.

Velocity v = Q / (W x H), using actual rectangular area. Circular equivalent diameter De = 1.30(WH)^0.625/(W+H)^0.25, with dimensions in consistent units. Darcy-Weisbach friction uses De and round equivalent velocity. Assumptions: rho=1.204 kg/m3, viscosity=1.81e-5 Pa s, galvanised-sheet roughness=0.09 mm.

Sizes use 50 mm increments with main/branch velocity limits 10/8 m/s, maximum friction 0.8 Pa/m and aspect ratio no greater than 4. The lowest compliant area is selected, with an aspect-ratio tie-break near 2:1. A lower achieved friction gradient is permitted where size increments or velocity govern.

The MIT Serghide correlation adapted from python-hvac was checked against iterative Colebrook and five ASHRAE Table 13 reference points. Four selected sections were also visually checked on the original ASHRAE 2021 Chapter 21 Figure 9 friction chart (Appendix C). The corresponding workshop reference is 2013 Figure 10. Both state density 1.20 kg/m3 and roughness 0.09 mm. Graphical readings are approximate; the numerical correlation controls exact sizing.

Full fitting, flexible-duct, VAV and AHU pressure losses have not been selected. This table is a straight-duct sizing calculation, not a complete fan selection. A complete system would require the critical-path loss sum and equipment pressure drops. The brief's required scope is supply layout and section sizing.

6. Verification and references

Native AutoCAD 2027 reopen confirms millimetre units and 80 two-square diffuser symbols, 20 VAV outlines, 80 flexible lines and 49 rigid section labels. The original source hashes remain unchanged. VAV and rigid-section flows reconcile with their downstream outlets. The current A1 PDF has been rendered and visually reviewed.

Additional geometric audit identified nine column footprints from source polylines and closed linework. Diffuser, VAV, rigid duct and flex footprints do not overlap these columns in plan. A 269 mm duplicated upper-branch route found in v03 is corrected in v04 as one common main section carrying both downstream VAV flows. No collinear rigid-centreline duplicates remain. This is a 2D audit, not a beam/headroom or fire approval.

Remaining issue work: finish detailed native geometry/label reconciliation, review any applicable marked drawing examples, and issue the final report/drawing package together. Ceiling-level and fire-strategy assumptions remain coordination notes. The present report does not declare final acceptance.

[1] Assignment brief, tasks 1-4; supplied architectural DWG and Layout PDF.

[2] Lecturer Week 8, Space air diffusion design and air duct sizing, pp.1, 4-5, 7 and 9; Canvas file 320901. Ductwork design and sizing, Canvas file 320903.

[3] Krueger published performance B1-12; Air Distribution Engineering a1-10/a1-11. Official PDFs preserved in 02_References.

[4] ASHRAE Fundamentals 2021 Chapter 21, Equation 25 and Tables 12-13; Applications 2015 Chapter 57 and 2019 Chapter 58 are comparison references, not substitutes for the recorded course example.

[5] FSD Circular 2/2023 Mechanical Ventilating Systems, sections 3, 7.1-7.2 and 9.2.

[6] EMSD BEC 2024, Table 6.4 and applicable fan-power/control provisions. No full BEC compliance claim follows from these supply-only calculations.

[7] TomLXXVI/python-hvac, MIT, commit 94247560a667905bfbf67e45a58965a58254fce8; licence and reviewed mathematical adaptation retained.

Exact source URLs and version decisions: 02_References/SOURCE_REGISTER.md. All numeric schedules: 05_Calculations/data/*_v04.csv. Reproduce with build_design_v04.py, draw_review_v04.py, verify_review_v04.py and build_report_v04.py.

Appendix A. Diffuser schedule (1-27)

All outlets: 600 mm panel / 8-inch neck. Q in L/s; T in m. Ratio ranges cover all four horizontal directions. P = pass against course 1.0-3.4 criterion, subject to Chapter 1 qualification. Full L values and pressure/NC estimates are in the CSV.

ID	Qmax	Qmin	Tmax	Tmin	Max T/L range	Min T/L range	Check
SD-01	152.2	60.9	4.45	2.68	2.12-2.80	1.28-1.69	P
SD-02	140.5	56.2	4.30	2.48	2.39-2.71	1.38-1.56	P
SD-03	140.5	56.2	4.30	2.48	2.39-2.71	1.38-1.56	P
SD-04	140.5	56.2	4.30	2.48	2.39-2.71	1.38-1.56	P
SD-05	140.5	56.2	4.30	2.48	2.39-2.71	1.38-1.56	P
SD-06	152.2	60.9	4.45	2.68	2.12-2.80	1.28-1.69	P
SD-07	152.2	60.9	4.45	2.68	2.12-2.80	1.28-1.69	P
SD-08	140.5	56.2	4.30	2.48	2.39-2.71	1.38-1.56	P
SD-09	140.5	56.2	4.30	2.48	2.39-2.71	1.38-1.56	P
SD-10	140.5	56.2	4.30	2.48	2.39-2.71	1.38-1.56	P
SD-11	140.5	56.2	4.30	2.48	2.39-2.71	1.38-1.56	P
SD-12	161.6	64.6	4.56	2.84	1.95-2.87	1.21-1.79	P
SD-13	161.7	64.7	4.56	2.84	2.17-2.53	1.35-1.58	P
SD-14	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-15	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-16	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-17	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-18	161.7	64.7	4.56	2.84	2.17-2.53	1.35-1.58	P
SD-19	161.7	64.7	4.56	2.84	2.17-2.53	1.35-1.58	P
SD-20	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-21	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-22	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-23	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-24	171.6	68.7	4.70	2.96	2.01-2.61	1.26-1.64	P
SD-25	148.2	59.3	4.40	2.62	2.09-2.93	1.25-1.74	P
SD-26	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-27	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P

Appendix A. Diffuser schedule (28-54)

All outlets: 600 mm panel / 8-inch neck. Q in L/s; T in m. Ratio ranges cover all four horizontal directions. P = pass against course 1.0-3.4 criterion, subject to Chapter 1 qualification. Full L values and pressure/NC estimates are in the CSV.

ID	Qmax	Qmin	Tmax	Tmin	Max T/L range	Min T/L range	Check
SD-28	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-29	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-30	148.2	59.3	4.40	2.62	2.09-2.93	1.25-1.74	P
SD-31	148.2	59.3	4.40	2.62	2.09-2.93	1.25-1.74	P
SD-32	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-33	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-34	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-35	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-36	157.3	62.9	4.51	2.77	1.93-3.00	1.18-1.85	P
SD-37	148.2	59.3	4.40	2.62	2.09-2.93	1.25-1.74	P
SD-38	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-39	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-40	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-41	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-42	148.2	59.3	4.40	2.62	2.09-2.93	1.25-1.74	P
SD-43	148.2	59.3	4.40	2.62	2.09-2.93	1.25-1.74	P
SD-44	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-45	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-46	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-47	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-48	157.3	62.9	4.51	2.77	1.93-3.00	1.18-1.85	P
SD-49	161.7	64.7	4.56	2.84	2.17-2.53	1.35-1.58	P
SD-50	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-51	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-52	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-53	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-54	161.7	64.7	4.56	2.84	2.17-2.53	1.35-1.58	P

Appendix A. Diffuser schedule (55-80)

All outlets: 600 mm panel / 8-inch neck. Q in L/s; T in m. Ratio ranges cover all four horizontal directions. P = pass against course 1.0-3.4 criterion, subject to Chapter 1 qualification. Full L values and pressure/NC estimates are in the CSV.

ID	Qmax	Qmin	Tmax	Tmin	Max T/L range	Min T/L range	Check
SD-55	161.7	64.7	4.56	2.84	2.17-2.53	1.35-1.58	P
SD-56	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-57	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-58	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-59	149.2	59.7	4.41	2.63	2.45-2.45	1.46-1.46	P
SD-60	171.6	68.7	4.70	2.96	2.01-2.61	1.26-1.64	P
SD-61	148.2	59.3	4.40	2.62	2.09-2.93	1.25-1.74	P
SD-62	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-63	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-64	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-65	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-66	148.2	59.3	4.40	2.62	2.09-2.93	1.25-1.74	P
SD-67	148.2	59.3	4.40	2.62	2.09-2.93	1.25-1.74	P
SD-68	136.8	54.7	4.26	2.42	2.37-2.84	1.35-1.61	P
SD-69	136.8	54.7	4.26	2.42	1.77-2.37	1.01-1.35	P
SD-70	136.8	54.7	4.26	2.42	1.77-2.37	1.01-1.35	P
SD-71	136.8	54.7	4.26	2.42	1.77-2.37	1.01-1.35	P
SD-72	157.3	62.9	4.51	2.77	1.88-2.50	1.15-1.54	P
SD-73	217.0	86.8	5.33	3.34	2.22-3.06	1.39-1.92	P
SD-74	217.0	86.8	5.33	3.34	2.22-3.06	1.39-1.92	P
SD-75	217.0	86.8	5.33	3.34	2.22-3.06	1.39-1.92	P
SD-76	217.0	86.8	5.33	3.34	2.22-3.06	1.39-1.92	P
SD-77	149.9	59.9	4.42	2.64	2.10-2.71	1.26-1.62	P
SD-78	149.9	59.9	4.42	2.64	2.10-2.71	1.26-1.62	P
SD-79	149.9	59.9	4.42	2.64	2.10-2.71	1.26-1.62	P
SD-80	149.9	59.9	4.42	2.64	2.10-2.71	1.26-1.62	P

Appendix B. Duct schedule (1-26)

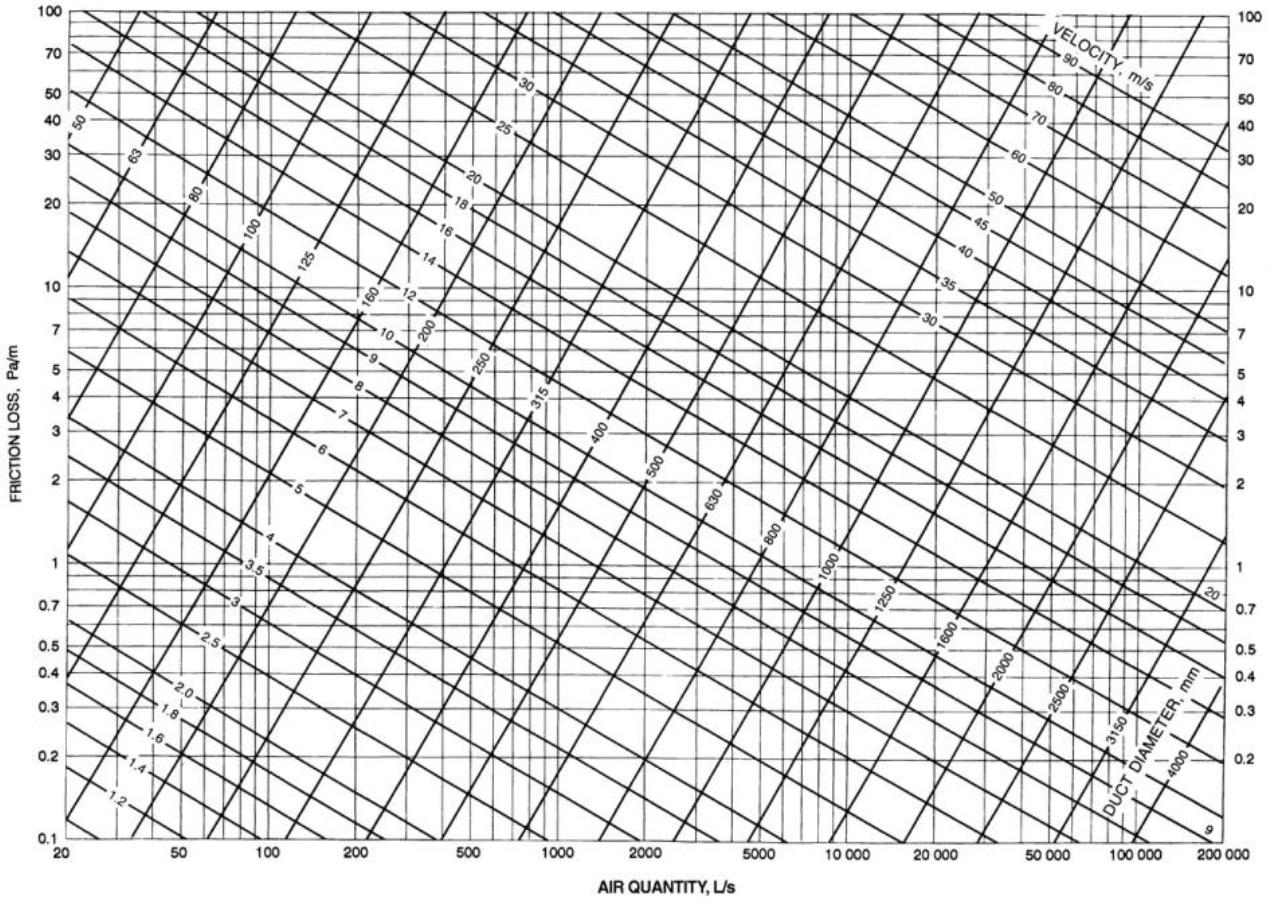
ID	Q L/s	W x H mm	Length m	m/s	Pa/m
M01	12000.4	1350 x 900	4.60	9.88	0.754
M02	5238.2	900 x 700	1.52	8.31	0.794
M03	3468.5	850 x 550	7.20	7.42	0.781
M04	1769.7	650 x 450	7.20	6.05	0.701
M05	6762.2	900 x 850	6.28	8.84	0.786
M06	4992.5	950 x 650	7.20	8.09	0.771
M07	3293.7	900 x 500	7.20	7.32	0.798
R13	1769.7	650 x 450	2.23	6.05	0.701
B13	595.9	350 x 350	0.80	4.86	0.776
R12	1173.7	450 x 450	6.90	5.80	0.792
B07	570.0	400 x 300	0.80	4.75	0.761
R11	603.7	350 x 350	6.90	4.93	0.795
B01	603.7	350 x 350	0.80	4.93	0.795
R23	1698.9	600 x 450	2.23	6.29	0.785
B14	572.1	400 x 300	0.80	4.77	0.767
R22	1126.8	500 x 400	6.90	5.63	0.762
B08	547.2	400 x 300	0.80	4.56	0.706
R21	579.6	400 x 300	6.90	4.83	0.785
B02	579.6	400 x 300	0.80	4.83	0.785
R33	1769.7	650 x 450	2.23	6.05	0.701
B15	595.9	350 x 350	0.80	4.86	0.776
R32	1173.7	450 x 450	6.90	5.80	0.792
B09	570.0	400 x 300	0.80	4.75	0.761
R31	603.7	350 x 350	6.90	4.93	0.795
B03	603.7	350 x 350	0.80	4.93	0.795
R43	1769.7	650 x 450	2.23	6.05	0.701

Appendix B. Duct schedule (27-49)

ID	Q L/s	W x H mm	Length m	m/s	Pa/m
B16	595.9	350 x 350	0.80	4.86	0.776
R42	1173.7	450 x 450	6.90	5.80	0.792
B10	570.0	400 x 300	0.80	4.75	0.761
R41	603.7	350 x 350	6.90	4.93	0.795
B04	603.7	350 x 350	0.80	4.93	0.795
R53	1698.9	600 x 450	2.23	6.29	0.785
B17	572.1	400 x 300	0.80	4.77	0.767
R52	1126.8	500 x 400	6.90	5.63	0.762
B11	547.2	400 x 300	0.80	4.56	0.706
R51	579.6	400 x 300	6.90	4.83	0.785
B05	579.6	400 x 300	0.80	4.83	0.785
R63	1826.3	650 x 450	2.23	6.24	0.743
B18	615.0	450 x 300	0.80	4.56	0.664
R62	1211.3	550 x 400	6.90	5.51	0.695
B12	588.3	350 x 350	0.80	4.80	0.758
R61	623.0	450 x 300	6.90	4.61	0.680
B06	623.0	450 x 300	0.80	4.61	0.680
M08	1467.4	550 x 450	1.72	5.93	0.736
M09	1467.4	550 x 450	4.70	5.93	0.736
M10	1467.4	550 x 450	0.27	5.93	0.736
B19A	733.7	500 x 300	6.98	4.89	0.721
B19B	733.7	500 x 300	0.97	4.89	0.721
B20B	733.7	500 x 300	0.97	4.89	0.721

Appendix C. Friction-chart cross-check

Original official ASHRAE chart, rotated for reading. Source: 2021 Fundamentals, Chapter 21 Figure 9. Four broad read-off intervals below are consistent with the calculated gradients; line thickness limits graphical precision.



Section	Q L/s	De mm	Chart Pa/m	Calc. Pa/m
M01	12000.4	1198.8	0.65-0.9	0.754
M04	1769.7	588.7	0.6-0.85	0.701
B01	603.7	382.6	0.65-0.95	0.795
B19A	733.7	420.0	0.6-0.85	0.721

Official chart and accompanying duct-design chapter. The four-point visual check supplements the full section-by-section numerical calculations; it is not claimed as exact chart digitisation.